

HQ router

Router#show ip route

Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP

D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area

N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2

E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP

i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area

* - candidate default, U - per-user static route, o - ODR

P - periodic downloaded static route

Gateway of last resort is 200.0.1.2 to network 0.0.0.0

10.0.0.0/28 is subnetted, 2 subnets

S 10.0.0.0 [1/0] via 200.0.2.2

S 10.0.0.16 [1/0] via 200.0.2.2

172.16.0.0/26 is subnetted, 3 subnets

C 172.16.0.0 is directly connected, FastEthernet0/0

C 172.16.0.64 is directly connected, FastEthernet6/0

C 172.16.0.128 is directly connected, FastEthernet1/0

192.168.1.0/27 is subnetted, 2 subnets

S 192.168.1.0 [1/0] via 200.0.1.2

S 192.168.1.32 [1/0] via 200.0.1.2

200.0.1.0/30 is subnetted, 1 subnets

C 200.0.1.0 is directly connected, Serial3/0

200.0.2.0/30 is subnetted, 1 subnets

C 200.0.2.0 is directly connected, Serial2/0

S* 0.0.0.0/0 [1/0] via 200.0.1.2

[1/0] via 200.0.2.2

```
Router#show ip protocols
Routing Protocol is "rip"
Sending updates every 30 seconds, next due in 18 seconds
Invalid after 180 seconds, hold down 180, flushed after 240
Outgoing update filter list for all interfaces is not set
Incoming update filter list for all interfaces is not set
Redistributing: rip
Default version control: send version 2, receive 2
  Interface          Send  Recv  Triggered RIP  Key-chain
FastEthernet0/0      22
FastEthernet1/0      22
FastEthernet6/0      22
Automatic network summarization is not in effect
Maximum path: 4
Routing for Networks:
  172.16.0.0
Passive Interface(s):
Routing Information Sources:
  Gateway            Distance           Last Update
Distance: (default is 120)
-
```

% Incomplete command.

Router#show ip nat translation

Pro	Inside global	Inside local	Outside local	Outside global
---	200.0.0.1	172.16.0.130	---	---

```
!  
interface FastEthernet0/0  
  ip address 172.16.0.1 255.255.255.192  
  ip nat inside  
  duplex auto  
  speed auto  
!  
interface FastEthernet1/0  
  ip address 172.16.0.129 255.255.255.192  
  ip nat inside  
  duplex auto  
  speed auto  
!  
interface Serial2/0  
  ip address 200.0.2.1 255.255.255.252  
  ip nat outside  
!  
interface Serial3/0  
  ip address 200.0.1.1 255.255.255.252  
  ip nat outside  
  clock rate 2000000  
!  
interface FastEthernet4/0  
  no ip address  
  shutdown  
!  
interface FastEthernet5/0  
  no ip address  
  shutdown  
!  
interface FastEthernet6/0  
  ip address 172.16.0.65 255.255.255.192  
  ip nat inside  
  duplex auto  
  speed auto  
!
```

```
!  
router rip  
  version 2  
  network 172.16.0.0  
  no auto-summary  
!  
ip nat pool p1 200.0.0.2 200.0.0.10 netmask 255.255.255.0  
ip nat inside source list 1 pool p1 overload  
ip nat inside source static 172.16.0.130 200.0.0.1  
ip classless  
ip route 0.0.0.0 0.0.0.0 200.0.1.2  
ip route 0.0.0.0 0.0.0.0 200.0.2.2  
ip route 192.168.1.0 255.255.255.224 200.0.1.2  
ip route 192.168.1.32 255.255.255.224 200.0.1.2  
ip route 10.0.0.0 255.255.255.240 200.0.2.2  
ip route 10.0.0.16 255.255.255.240 200.0.2.2  
!  
ip flow-export version 9  
!  
!  
access-list 1 permit 172.16.0.0 0.0.0.63  
access-list 1 permit 172.16.0.64 0.0.0.63  
!
```

Router#

Router#show run | include nat

ip nat inside

ip nat inside

ip nat outside

ip nat outside

ip nat inside

ip nat pool p1 200.0.0.2 200.0.0.10 netmask 255.255.255.0

ip nat inside source list 1 pool p1 overload

ip nat inside source static 172.16.0.130 200.0.0.1

Router#

B2 router


```
Router#show ip route
```

```
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
```

```
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
```

```
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
```

```
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
```

```
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
```

```
* - candidate default, U - per-user static route, o - ODR
```

```
P - periodic downloaded static route
```

```
Gateway of last resort is 200.0.2.1 to network 0.0.0.0
```

```
10.0.0.0/28 is subnetted, 2 subnets
```

```
C      10.0.0.0 is directly connected, FastEthernet1/0
```

```
C      10.0.0.16 is directly connected, FastEthernet0/0
```

```
200.0.2.0/30 is subnetted, 1 subnets
```

```
C      200.0.2.0 is directly connected, Serial2/0
```

```
S* 0.0.0.0/0 [1/0] via 200.0.2.1
```

```
Router#show ip protocols
Routing Protocol is "rip"
Sending updates every 30 seconds, next due in 7 seconds
Invalid after 180 seconds, hold down 180, flushed after 240
Outgoing update filter list for all interfaces is not set
Incoming update filter list for all interfaces is not set
Redistributing: rip
Default version control: send version 2, receive 2
  Interface          Send  Recv  Triggered RIP  Key-chain
  FastEthernet0/0     22
  FastEthernet1/0     22
  Serial2/0           22
Automatic network summarization is not in effect
Maximum path: 4
Routing for Networks:
  10.0.0.0
  200.0.2.0
Passive Interface(s):
Routing Information Sources:
  Gateway            Distance          Last Update
Distance: (default is 120)
```

```
!  
interface FastEthernet0/0  
  ip address 10.0.0.17 255.255.255.240  
  duplex auto  
  speed auto  
!  
interface FastEthernet1/0  
  ip address 10.0.0.1 255.255.255.240  
  duplex auto  
  speed auto  
!  
interface Serial2/0  
  ip address 200.0.2.2 255.255.255.252  
  clock rate 2000000  
!  
interface Serial3/0  
  no ip address  
  shutdown  
!  
interface FastEthernet4/0  
  no ip address  
  shutdown  
!  
interface FastEthernet5/0  
  no ip address  
  shutdown  
!  
router rip  
  version 2  
  network 10.0.0.0  
  network 200.0.2.0  
  no auto-summary  
!  
ip classless  
ip route 0.0.0.0 0.0.0.0 200.0.2.1  
!  
ip flow-export version 9  
!  
!
```

B1 router

```
Router#show ip route
```

```
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
```

```
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
```

```
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
```

```
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
```

```
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
```

```
       * - candidate default, U - per-user static route, o - ODR
```

```
       P - periodic downloaded static route
```

```
Gateway of last resort is 200.0.1.1 to network 0.0.0.0
```

```
    192.168.1.0/27 is subnetted, 2 subnets
```

```
C      192.168.1.0 is directly connected, FastEthernet0/0
```

```
C      192.168.1.32 is directly connected, FastEthernet1/0
```

```
    200.0.1.0/30 is subnetted, 1 subnets
```

```
C      200.0.1.0 is directly connected, Serial2/0
```

```
S*    0.0.0.0/0 [1/0] via 200.0.1.1
```

```
Router#show ip protocols
Routing Protocol is "rip"
Sending updates every 30 seconds, next due in 26 seconds
Invalid after 180 seconds, hold down 180, flushed after 240
Outgoing update filter list for all interfaces is not set
Incoming update filter list for all interfaces is not set
Redistributing: rip
Default version control: send version 2, receive 2
  Interface          Send  Recv  Triggered RIP  Key-chain
  FastEthernet0/0      22
  FastEthernet1/0      22
  Serial2/0            22
Automatic network summarization is not in effect
Maximum path: 4
Routing for Networks:
  192.168.1.0
  200.0.1.0
Passive Interface(s):
Routing Information Sources:
  Gateway            Distance          Last Update
Distance: (default is 120)
```

```
!  
interface FastEthernet0/0  
  ip address 192.168.1.1 255.255.255.224  
  duplex auto  
  speed auto  
!  
interface FastEthernet1/0  
  ip address 192.168.1.33 255.255.255.224  
  duplex auto  
  speed auto  
!  
interface Serial2/0  
  ip address 200.0.1.2 255.255.255.252  
!  
interface Serial3/0  
  no ip address  
  clock rate 2000000  
!  
interface FastEthernet4/0  
  no ip address  
  shutdown  
!  
interface FastEthernet5/0  
  no ip address  
  shutdown  
!  
router rip  
  version 2  
  network 192.168.1.0  
  network 200.0.1.0  
  no auto-summary  
!  
ip classless  
ip route 0.0.0.0 0.0.0.0 200.0.1.1  
!  
ip flow-export version 9  
!
```